one temporal order. They never happen in reverse, and so they provide a notion of before and after—they give us a consistent and seemingly universal conception of past and future. These observations convince us that were we to examine all of spacetime from the outside (as in Figure 5.1), we would see significant asymmetry along the time axis. Splattered eggs the world over would lie to one side—the side we conventionally call the future—of their whole, unsplattered counterparts.

Perhaps the most pointed example of all is that our minds seem to have access to a collection of events that we call the past—our memories—but none of us seems able to remember the collection of events we call the future. So it seems obvious that there is a big difference between the past and the future. There seems to be a manifest orientation to how an enormous variety of things unfold in time. There seems to be a manifest distinction between the things we can remember (the past) and the things we cannot (the future). This is what we mean by time's having an orientation, a direction, or an arrow.[1]

Physics, and science more generally, is founded on regularities. Scientists study nature, find patterns, and codify these patterns in natural laws. You would think, therefore, that the enormous wealth of regularity leading us to perceive an apparent arrow of time would be evidence of a fundamental law of nature. A silly way to formulate such a law would be to introduce the Law of Spilled Milk, stating that glasses of milk spill but don't unspill, or the Law of Splattered Eggs, stating that eggs break and splatter but never unsplatter and unbreak. But that kind of law buys us nothing: it is merely descriptive, and offers no explanation beyond a simple observation of what happens. Yet we expect that somewhere in the depths of physics there must be a less silly law describing the motion and properties of the particles that make up pizza, milk, eggs, coffee, people, and stars—the fundamental ingredients of everything—that shows why things evolve through one sequence of steps but never the reverse. Such a law would give a fundamental explanation to the observed arrow of time.

The perplexing thing is that no one has discovered any such law. What's more, the laws of physics that have been articulated from Newton through Maxwell and Einstein, and up until today, show a *complete symmetry between past and future*.[10] Nowhere in any of these laws do